

LESSON 136 GROUND PORTION**Lesson Objective**

The purpose of this lesson is to evaluate the student's ability to explain all elements listed in the oral evaluation.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - The student was assigned to study in preparation for this lesson.
- Student questions.

References

- Aeronautical Information Manual.
 - Chapter 01 - Air Navigation.
 - Chapter 03 - Airspace.
- Federal Aviation Regulations
 - Part 01.
 - Part 61.
 - Part 91.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 03 - Aircraft Construction.
 - Chapter 06 - Flight Controls.
 - Chapter 07 - Aircraft Systems.
- Warrior Cadet/III Pilot's Information Manual.
 - Section 02 - Limitations.
 - Section 04 - Normal Procedures.
 - Section 05 - Performance.
 - Section 06 - Weight and Balance.
 - Section 07 - Systems.

Lesson Content**V Speeds and Emergencies Quiz**

- There is no need to administer the V speeds and emergencies quiz for this lesson. However, use the student's TOLD card to quiz them on the V speeds.
- At this stage, the student must be able to accurately describe all of the V speeds.

Flight Operations Manual

- Quiz the student on the following items from the FOM:
 - Solo XC weather minimums.
 - 7SM and 2,500ft.
 - Solo students are not permitted to take off or land with a tailwind of any velocity.
 - Cross-country flights.

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- Must receive a weather briefing.
- File and open a flight plan.
- Student pilots must complete all landings to a full-stop/full-stop taxi back when solo.
- Solo student pilots on a cross-country flight must return to Melbourne no later than 30 minutes before official sunset.
- Fuel requirements.
 - 1 hour of fuel reserve at all times.
 - PA28-161/181 departures are not authorized unless at least one tank is filled to the tabs.
 - Cross-Country departures must carry the maximum fuel allowed by weight and balance calculations.
 - Flights exceeding 3.5 hours Hobbs time require a fuel stop.

Operation of Systems

- Carburetor icing:
 - Give the student a scenario where the engine is running rough in a high humidity and low-temperature environment.
 - See if the student can correlate this scenario to carburetor icing.
 - Ensure that the student can run through the correct procedure.
- Low fuel situation:
 - Review situations that might lead to low fuel.
 - Leak, poor preparation, mixture too rich, deviating from POH, headwinds, etc.
 - Ask the student what they would do in case of a low-fuel situation.
 - Divert, lean for the best economy, declare an emergency, etc.
- Ask general system questions to ensure the student has been actively reviewing (don't spend too much time in this section, just enough to ensure the student is still proficient).

Federal Aviation Regulations

- Pilot in command (PIC) responsibilities - 91.3:
 - Since the student will be PIC during the solo cross-countries, quiz them on what they are responsible for as a PIC.
 - Airworthiness, safety of flight, final authority.
- Preflight action - 91.103:
 - What preflight action does a flight not in the vicinity of an airport require?
 - NOTAMs.
 - Weather.
 - Known delays.
 - Runway lengths.
 - Alternates if needed.
 - Fuel requirements.
 - Takeoff/landing data.

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- Minimum safe altitudes - 91.119:
 - Quiz the student on the minimum safe altitudes:
 - Congested → 1,000 ft above, 2,000 ft horizontally.
 - Other than congested → 500 ft above.
 - Sparsely populated and open water → cannot operate within 500 feet of the nearest vessel, structure, person, or vehicle.
- FAA fuel reserves - 91.151:
 - What are the FAA fuel reserves? Does FIT have the same fuel reserve requirements?
 - FAA
 - Day - 30 mins.
 - Night - 45 mins.
 - FIT
 - Always - 60 mins.
- VFR cruising altitudes - 91.159:
 - What is the purpose? What does the regulation say?
 - The purpose is for separation on converging courses.
 - Based on the magnetic course starting above 3,000 ft!
 - 0-179 degrees → odd 1000's + 500.
 - 180 - 359 degrees → even 1000's + 500.
- Required endorsements:
 - Required pre-solo endorsements - 61.87.
 - Pre-solo aeronautical knowledge test - 61.87(b).
 - Pre-solo flight training - 61.87(c).
 - Solo flight (initial 90 days) - 61.87(n).
 - Require solo cross-country endorsements - 61.93.
 - Cross-country flight training - 61.93(c)1(c)2.
 - Solo cross-country flight (specific route) - 61.93(c)3.

Airworthiness Requirements

- Don't spend too much time in this section, as this was previously assessed on the stage 1 check.
- Ask airworthiness questions pertinent to the planned cross-country:
 - What are the required equipment and documents needed to fly this cross-country?
 - Does the aircraft meet all of the required inspections? What are the required inspections?
 - What would you do if you got to the airplane and realized something was missing or inoperative?
 - Make sure they can run through the TARK acronym.

Cross-country Flight Planning

- Inspect the nav log, sectional chart, and TOLD card before leading into scenarios.
- Ensure all boxes are filled in correctly and appropriately.
- Ensure that the TOLD card fuel calculations match what is on the nav log.
- Airport information:

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- Have the student brief information from the chart supplement:
 - Frequencies.
 - Runway lengths.
 - Fuel services.
 - Any LAHSO information.
 - Airport lighting.
 - Remarks.
- Navigation log:
 - Make sure the flight plan form is filled out correctly.
 - Ask the student why the cruising altitude was chosen - 91.159.
 - Ask the student why it is a safe altitude - 91.119.
 - Evaluate fuel, time, and distance totals.
 - Ask if TAS increases with wind.
 - Discuss the effects of density altitude on TAS.
 - Ask the student what information was required to plan this flight - 91.103.
 - Discuss cruising performance values:
 - Top of climb.
 - Fuel requirement – discuss mixture leaning techniques.
 - Best economy vs best power.
 - How to perform both.
 - Engine RPM.
 - True airspeed.
 - Top of descent.
 - Takeoff/landing data.
 - Range and endurance, what's the difference?
- NOTAMs:
 - Ensure that the student has the applicable NOTAMs on the nav log.
 - Quiz the student on the different types of NOTAMs.
 - Types (AIM 5-1-3):
 - NOTAM (D).
 - FDC NOTAM.
 - International NOTAM.
 - Military NOTAM.
- Pilotage and dead reckoning:
 - What is pilotage? What is dead reckoning?
 - Pilotage: navigation by reference to landmarks or checkpoints.
 - Dead reckoning: navigation solely by computation based on time, airspeed, distance, and direction.

Flight Plan

- Review the student's flight plan.
- Ensure that all of the required information is filled out.
- How was the enroute endurance calculated?
- What does SG/LB1 mean in the equipment section?
 - S: VOR, VHF, ILS.
 - G: GNSS
 - L: Mode S transponder (ID, position, altitude, data exchange).
 - B1: ADS-B OUT

Airspace

- Check for airspace recognition on the sectional chart.
- Quiz the student on chart symbology.
- Ask the student about the types of airspace
 - Discuss weather minimums where appropriate.
- Controlled and uncontrolled airspace - 91.127 through 91.135:
 - Class A:
 - What is class A airspace?
 - Generally, from FL180 to FL600.
 - Extends 12 NM offshore of 48 contiguous states and Alaska.
 - What are the entry requirements and required equipment?
 - Entry requirements:
 - ATC Clearance.
 - IFR flight plan.
 - Required equipment:
 - Two-way radios.
 - Mode C transponder.
 - ADS B-OUT.
 - What are the weather minimums?
 - None, IFR.
 - Class B:
 - What is class B airspace?
 - Generally, from SFC to 10,000 ft MSL.
 - Busy airports: based on total operations or passenger enplanements.
 - Multiple layers (upside-down cake).
 - Configuration can be different depending on the airport.
 - ATC provides separation within the airspace.
 - Some airports do not allow student, sport, or recreational pilots to land, usually due to high traffic volume
 - Part 91, appendix D, section 04.
 - What are the entry requirements and required equipment?

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- Entry requirements:
 - ATC Clearance.
- Required equipment:
 - Two-way radios.
 - Mode C transponder (in class B, above and within the lateral limits of class B, and within 30 nm (mode C veil)).
 - ADSB-OUT (in class B, above and within the lateral limits of class B, and within 30 nm (mode C veil)).
- What are the weather minimums?
 - 3SM visibility and clear of clouds.
- Class C:
 - What is class C airspace?
 - Generally, from SFC to 4,000 ft MSL.
 - Surrounds airports that have a control tower, radar approach control, and have a certain number of IFR operations or passenger enplanements.
 - Configuration can be different depending on the airport.
 - Two layers:
 - Inner ring extends from the SFC to 4,000 ft (typically) with a 5 NM radius.
 - Outer ring extends from 1,200 ft to 4000 ft (typically) with a 10 NM radius.
 - What are the entry requirements and required equipment?
 - Entry requirements:
 - Establish two-way radio communication.
 - Required equipment:
 - Two-way radios.
 - Mode C transponder (in class C and above and within the lateral limits of class C)
 - ADSB-OUT (in class C and above and within the lateral limits of class C).
 - What are the weather minimums?
 - 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
- Class D:
 - What is class D airspace?
 - Generally, from SFC to 2,500 ft MSL.
 - Surrounds airports that have a control tower.
 - Configuration can be different depending on the airport (KMLB!)
 - Usually designed to contain instrument approaches.
 - What are the entry requirements and required equipment?
 - Entry requirements:

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- Establish two-way radio communication.
- Required equipment:
 - Two-way radios.
- What are the weather minimums?
 - 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
- Class E:
 - What is class E airspace?
 - Controlled airspace that is not class A, B, C, or D.
 - Extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
 - Sectional and other charts depict all locations of Class E airspace with bases below 14,500 feet MSL. In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.
 - In most areas, the Class E airspace base is 1,200 feet AGL. In many other areas, the Class E airspace base is either the surface or 700 feet AGL.
 - All airspace above FL 600 is Class E airspace.
 - What are the entry requirements and required equipment?
 - ADSB-OUT: at or above 10,000 feet MSL, excluding airspace at and below 2,500 feet AGL; over the Gulf of Mexico, at and above 3,000 feet MSL, within 12 nm of the U.S. coast.
 - Mode C transponder: at and above 10,000 feet MSL, excluding the airspace at and below 2,500 ft AGL.
 - What are the weather minimums?
 - Less than 10,000ft MSL: 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
 - More than 10,000ft MSL: 5SM visibility and 1,000ft above, 1,000ft below, 1SM horizontally.
- Class G:
 - What is class G airspace?
 - Uncontrolled.
 - ATC has no authority or responsibility.
 - Although ATC has no authority, a pilot cannot be careless or reckless as this would break the 91.13 regulation.
 - VFR rules apply.
 - What are the entry requirements and required equipment?
 - None.
 - What are the weather minimums?
 - Less than 1,200ft AGL:
 - Day: 1SM visibility and clear of clouds.
 - Night: 3SM visibility and 1,000 ft above, 500 ft below, 2000ft

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- horizontally.
- More than 1,200ft AGL and less than 10,000ft MSL:
 - Day: 1SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
 - Night: 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
- More than 10,000ft MSL:
 - 5SM visibility and 1,000ft above, 1,000ft below, 1SM horizontally.
- Special use airspace:
 - What are the different types of special use airspace?
 - **Warning area.**
 - The government does not have jurisdiction over airspace.
 - Defined dimensions, extending 3NM offshore of the U.S.
 - Contains activity that may be hazardous to nonparticipating aircraft.
 - Charted as “W” followed by a number.
 - **Alert area.**
 - High volume of pilot training or an unusual type of aerial activity.
 - Depicted with an “A” followed by a number on charts.
 - **Restricted area.**
 - Area in which operations are hazardous to non-participating aircraft.
 - Artillery firing, guided missiles.
 - Flight of aircraft may be restricted, but not prohibited.
 - Must have permission from ATC or the controlling agency to enter.
 - Charted with an “R” followed by a number.
 - Information is found on the back of charts.
 - **Military operation area.**
 - Defined vertical and lateral limits.
 - Separates certain military training activities from IFR traffic.
 - IFR traffic may be cleared through airspace if ATC can provide separation.
 - Additional information for each is found on the back of the sectional.
 - **Prohibited area.**
 - Airspace in which flight is prohibited.
 - Camp David, White House, Congressional buildings.
 - Usually for security or national welfare.
 - Charted at “P” followed by a number.
 - **Controlled firing area.**
 - Contains activity that is hazardous to non-participating aircraft.
 - Activity must stop if the spotter spots non-participating aircraft.
 - Published in chart supplement under Special Notices.

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- Other airspace:
 - What are some other types of airspace?
 - Military Training Routes (MTRs):
 - Used by military aircraft to maintain proficiency in tactical flying.
 - Usually established below 10,000 feet MSL for operations exceeding 250 knots.
 - Routes depicted as IR (if IFR) and VR (if VFR), followed by a number:
 - Those not exceeding 1,500 feet AGL contain 4 numbers (IR 1206).
 - Those above 1,500 feet AGL contain 3 numbers (VR 207).
 - Parachute jump aircraft operations:
 - Airports at which there are high numbers of parachute operations.
 - Published in chart supplement under Parachute Jumping Areas and depicted on sectional charts.
 - Published VFR Routes
 - Used for transitioning around, under, or through complex airspace
 - Usually found on VFR terminal area planning charts
 - Often referred to as VFR flyway, VFR corridor, Class B airspace VFR transition route, and terminal area VFR route
 - Terminal Radar Service Areas (TRSAs)
 - Areas in which pilots can receive additional radar services.
 - Purpose is to provide separation for IFR and participating VFR traffic.
 - The airspace within the TRSA becomes Class D around the airports.
 - Remaining portion overlies other controlled airspace, which is normally Class E.
 - Depicted on VFR sectional charts and terminal area charts with a solid black line and altitudes for each segment.
 - Participation is voluntary and encouraged.
 - Temporary flight restrictions (TFRs)
 - Designated by flight data center (FDC) Notice to Airmen (NOTAM)
 - Begins with phrase “FLIGHT RESTRICTIONS” followed by location, effective time period, and defined in statute miles, and altitudes affected.
 - Purposes may include:
 - Protect persons or property in air or on surface from existing or imminent hazard.
 - Provide a safe environment for operation of disaster relief aircraft.
 - Prevent unsafe congestion of sightseeing aircraft above incident or event.

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- Protect declared national disasters for humanitarian reasons in Hawaii.
- Protect the President, Vice President, or other public figures.
- Provide a safe environment for space agency operations.
- If penetrated, the pilot may undergo security investigations and certificate suspensions.

Weather

- Weather briefings:
 - Ask the student to go through the weather briefing they obtained.
 - What type of briefing did they obtain? What are the different types of briefings?
 - Outlook.
 - Standard.
 - Abbreviated.
 - In-flight.
 - What constitutes an official weather briefing?
 - Official brief: credentials and tail number (ex: 1800wxbrief, FSS, ForeFlight).
 - Non-official brief: aviationweather.gov, weather apps...
- METARs:
 - Quiz the students on metar valid times.
 - Pull up sample METARs with less common reports on them to see if the student knows or can figure out what is being observed.
- TAF ([AC 00-45H](#)):
 - Quiz the student on TAF valid times.
 - Pull up sample TAFs with less common reports on them to see if the student knows or can figure out what is being forecast.
- GFA tool:
 - What is the GFA tool?
 - What does it show?
- Winds and temperature aloft chart:
 - How often is it issued?
 - 4 times a day (3 times a day on AWC).
 - How long is it valid?
 - Shown on the chart.
 - What does it include?
 - Winds and temps at specific locations.
 - The wind is in true knots.
 - The temperature is in Celsius.
 - How does it code the winds if they are forecasted to be 100-199 knots?
 - Add 50 to the wind direction and subtract 100 from the speed.
 - Show an example to see if the student can decode it.

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- How does it code the winds if they are forecasted to be calm or less than 5 knots?
 - 9900.
- How does it code the winds if they are forecasted to be 200 knots or greater?
 - 7799: 270 degrees and greater than 199 knots.
- Why do some stations not report winds or temperature?
 - No winds are forecasted within 1,500 feet of station elevation.
 - No temperatures are forecasted within 2,500 AGL of station elevation.
- Radar imagery:
 - What are the different types of radar?
 - National radar:
 - Shows the intensity of the precipitation.
 - Provides a single view of the entire country.
 - Updated every 10 minutes.
 - Loops the latest 2 hours of images.
 - Single site radar:
 - Composed of 140 radar sites across the U.S.
 - Each site shows base reflectivity, composite reflectivity, and total precipitation.
 - Updated every 10 minutes.
 - Loops the latest 2 hours of images.
- Significant weather prognostic chart:
 - What are the different types? What do they show? How often are they issued?
 - Low level significant weather - surface to FL240.
 - High level - FL250 to FL600.
 - 2 forms:
 - 12 and 24 hour forecast chart:
 - Issued 4 times a day.
 - Valid time printed on chart.
 - Shows forecasted weather to include non convective turbulence, freezing levels, and IFR or MVFR weather.
 - Cover the difference between IFR, MVFR, and VFR.
 - IFR: less than 1000ft ceilings and less than 3SM visibility.
 - MVFR: 1000ft - 3000ft ceiling and 3-5SM visibility.
 - VFR: more than 3000ft ceiling and more than 5SM visibility.
 - 36 and 48-hour surface forecast chart:
 - Extension of above chart.
 - Surface weather forecasts.
 - Issued 2 times a day.
 - Pressure patterns, fronts, and precipitation.

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- Surface analysis chart:
 - How often are they issued? How long are they valid for? What do they show?
 - Chart depicts the distribution of several items including sea level pressure, the position of the highs, lows, ridges, and troughs, the location and type of fronts, wind direction and speed, temperature and dew point.
 - Analysis of current weather.
 - Computerized.
 - Transmitted every 3 hours.
 - Valid time shown on chart (observed data).
 - What are high- and low-pressure systems?
 - In the northern hemisphere – flow of air from high to low pressure is deflected to the right due to the Coriolis force.
 - High pressure systems (outward - downward - clockwise).
 - Dry, stable, descending air.
 - Good weather.
 - Low pressure systems (inward - upward - counterclockwise).
 - Air flows in a low-pressure area to replace rising air.
 - Bad weather.
 - What is a cold front?
 - Occurs when a mass of cold, dense and stable air advances and replaces a body of warmer air.
 - Moves at 25-30 mph, can move as fast as 60 mph.
 - Rapidly ascending air causes the temperature to decrease suddenly, forcing creation of clouds.
 - It “lifts moist air and creates instability” - recipe for thunderstorms.
 - Prior to cold front passing, cirriform towering cumulus clouds are present, and cumulonimbus clouds may develop.
 - Showery precipitation.
 - After frontal passage, clouds dissipate to cumulus clouds with a corresponding decrease in a precipitation
 - Good visibility prevails and winds from the west and northwest
 - Flight toward a cold front:
 - Flight progresses to the oncoming cold front; the clouds show signs of vertical development with a broken layer.
 - Pilots should use sound judgment on frontal passage.
 - What is a warm front?
 - Occurs when the warm mass of air advances and replaces a body of colder air. Warm fronts move slowly, 10-25 mph. Slope of the advancing front slide over the top of the cooler air and gradually pushes it out of the area. As the warm air is lifted, the temperature drops, and condensation occurs from the humidity of the air.

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- Generally, cirriform, stratiform clouds (stable air) along with fog can be expected to form along the frontal boundary. (rain evaporates when it gets close to the cooler ground).
- Light to moderate steady precipitation.
- Flight toward a warm front:
 - Visibility decreases.
 - Falling barometric pressure.
 - Weather overall deteriorates.
 - May be impossible to continue a VFR flight after flying through a warm front.
- What is a stationary front?
 - When the forces of two air masses are relatively equal, the boundary layer or front that separates them remains stationary and influences the local weather for days.
- What is an occluded front?
 - Fast moving cold front catches up with a slow-moving warm front. Warm front weather prevails but is followed by cold front weather.
- What is frontogenesis? What about frontolysis?
 - Frontogenesis: the initial formation of a front.
 - Frontolysis: the dissipation of a front.
- Pilot reports (PIREPs):
 - What are the different types of PIREPs? What is included in a PIREP?
 - Weather report from a flight deck. Consists of message type, location, time, altitude/flight level, type aircraft, and at least one other element to describe the report phenomena.
 - 2 types: routine (UA) and urgent (UUA):
 - Urgent includes:
 - Tornadoes.
 - Severe or extreme turbulence.
 - Hail.
 - Low level wind shear within 2000 ft.
 - Contains information regarding conditions as they actually exist.
 - When must you submit a PIREP?
 - When requested by another pilot.
 - When the ceiling is at 5,000ft or less.
 - When visibility is 5SM or less.
 - Thunderstorms.
 - Moderate or greater turbulence.
 - Light or greater icing.
 - Wind shea.
 - Volcanic ash.
 - Show the student various PIREPs and see if they can decode them properly.

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- AIRMETs, SIGMETs, and convective SIGMETs:
 - What is an AIRMET (WA)?
 - Light to moderate weather that may be hazardous to aircraft (primarily light).
 - Issued every 6 hours Valid for 6.
 - Sierra
 - IFR conditions (ceilings less than 1000 ft and vis less than 3sm and mountain obscurations.
 - Tango
 - Light to moderate Turbulence.
 - surface winds 30 knots or greater.
 - Low-level wind shear.
 - Zulu
 - Icing.
 - Freezing levels.
 - An airmet will be issued if any of these conditions are expected to happen over an area of at least 3,000 square miles.
 - What is a SIGMET (WS)?
 - Severe weather conditions that may be hazardous to aircraft.
 - Unscheduled (comes out when needed).
 - Valid for 4 hours (6 for hurricanes).
 - Severe icing not associated with thunderstorms.
 - Severe or extreme turbulence or CAT not associated with thunderstorms.
 - Dust/sandstorms.
 - Volcanic ash.
 - Issued with alphabetic identifier (no sierra or tango) → different zones throughout the USA have designated identifiers.
 - Issued when these weather conditions are forecasted to happen over an area of at least 3000 square miles or an area judged to have a significant impact on safety of aircraft.
 - What is a convective SIGMET (WST)?
 - Hazardous convective weather affecting every flight.
 - Issued at 55 minutes past the hour
 - Valid for 2 hours
 - Severe thunderstorms with surface winds greater than 50 knots
 - Hail at surface at least $\frac{3}{4}$ inch in diameter
 - Affects 40% or more of a 3000 sq miles area.
 - Line of thunderstorms at least 60 miles long with thunderstorms affecting at least 40 percent of its length.
 - Tornadoes

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- Numbered sequentially depending on the zone (W).
 - If no convective weather, will say “NONE” Est, Central, East
- Center weather advisory:
 - What is a center weather advisory? Who issues them? How long are they valid for?
 - A CWA is an aviation weather warning for conditions meeting or approaching national in-flight advisory (AIRMET, SIGMET, or Convective SIGMET) criteria.
 - CWAs are valid for up to 2 hours and may include forecasts of conditions expected to begin within 2 hours of issuance.
- Convective outlook:
 - What is the use of a convective outlook?
 - Consist of both a narrative and a graphic showing severe thunderstorm threats across the US.
 - Convective outlooks are split into 4 different charts:
 - Day 1: risk of severe weather for 24 hours. Issued 5 times a day.
 - Day 2: risk of severe weather for the next 24 hours. Issued 2 times a day.
 - Day 3: risk of severe weather for the next 24 hours. Issued daily.
 - Day 4-8: risk of severe weather for day 4 through day 8. Issued daily.
- AWOS, ASOS, ATIS:
 - What is the difference between AWOS, ASOS, and ATIS?
 - Check the chart supplement to see what type of weather system is in use.
 - AWOS:
 - Automated Weather Observing System.
 - Operated and controlled by the FAA.
 - Automated and reports the weather every minute.
 - Types of AWOS:
 - AWOS-A: Altimeter setting.
 - AWOS-AV: Altimeter and visibility.
 - AWOS-I: Altimeter, wind, temperature, dew point, and density altitude.
 - AWOS-II: AWOS-I plus visibility.
 - AWOS-III: AWOS-II plus ceiling.
 - AWOS-III P: AWOS-III plus precipitation identification.
 - AWOS-III T: AWOS-III plus thunderstorm/lightning identification.
 - AWOS-III PT: AWOS-III plus precipitation and thunderstorm/lightning identification.
 - AWOS-IV: AWOS-III plus precipitation, type and accumulation, freezing rain, thunderstorm, and runway surface sensors.
 - ASOS:

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- Automated Surface Observing System.
 - Operated and controlled by the National Weather Service (NWS), Department of Defense (DoD), and sometimes by the FAA.
 - Used for the national weather system, not just for aviation purposes.
 - Reported information:
 - Wind speed and direction.
 - Visibility.
 - Sky condition.
 - Ceiling height and precipitation.
 - Barometric pressure.
 - Density altitude.
- ATIS:
 - Automated Terminal Information Service.
 - Towered airports.
 - Requires a human to monitor the ATIS.
 - ATIS information is obtained from the ASOS system.
 - Contains more than just weather (NOTAMs, remarks, etc...)
 - Published hourly at 55 minutes past the hour.
 - If the weather is rapidly changing, the tower can give an updated ATIS (like a SPECI METAR).
 - Reported information:
 - Phonetic alphabet letter.
 - Time of the report.
 - Wind speed and direction.
 - Visibility.
 - Cloud cover.
 - Temperature.
 - Dew point.
 - Barometric pressure.
 - NOTAMs, remarks, other information.

Lost Procedures

- Give the student a scenario where they find themselves in a lost situation.
- Ensure the student can list out what actions they would take:
 - 7C's:
 - Calm.
 - Climb.
 - Identify prominent landmarks.
 - Circle.
 - Avoid aggravating the situation.
 - Crosscheck.

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- GPS.
- VOR: use 2 VOR's, center the CDI, draw on the sectional chart to determine position.
- Communicate.
 - Request assistance from ATC.
- Comply.
 - Follow ATC instructions.
- Conserve.
 - Reduce power to 55% and lean the mixture for best economy (refer to PIM).

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 136.
- Assign homework:
 - Review maneuvers to be performed in 136.
 - Plan the cross-country route of the flight to be flown in 136.

Completion Standards

This lesson is complete when the student demonstrates the ability to explain all elements listed in the oral evaluation.